Lecture 02_Social Context

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2.1 A Very Short History of IT

- It was during the **Second World War**, and in its aftermath, that major technological breakthroughs in electronics took place: the first programmable computer and the transistor.

- **1970s** - new information technologies became widely diffused, accelerating their synergistic development and converging into a new paradigm.
- These occurred in several stages of innovation in the *three main technological fields*: **micro-electronics**, **computers**,and **telecommunications**.

Transistor

- The **transistor** made the fast processing of electric impulses in a binary mode possible.
- This enabled the **coding of logic and communication between machines.**
- **Semiconductor** processing devices—integrated circuits or ‘chips’—are now made of millions of transistors.

Transistor

- The comparatively ‘giant leap’ forward occurred in **1971** when Intel introduced the 4-bit 4004 microprocessor, that is the computer on a chip, and information-processing power could thus be installed everywhere (where the buyer had enough resources).

- In the last two decades of the **20th century** and the first decade of the **21st**, increasing chip power resulted in a dramatic enhancement of micro-computing power.
- Since the **mid-1980s**, microcomputers could no longer be conceived of in isolation: they were linked up in networks, with increasing mobility, on the basis of portable computers (and later mobile phones).

- Into the **2010s**, storage capacity was so cheap and computing power increased enough, that the time of **Big Data** commenced, where massive amounts of data are analyzed algorithmically to find patterns.
- Such data may have been collected on purpose for it, such as the measurements to compute the black hole image and brain scan image analyses, or ‘on the side’ initially and then exploited and collected on purpose, such as online user behavior.

Big Data

- Big Data has no single definition, but there are either ***3 Vs or 5 Vs*** associated with it: **Volume**, **Velocity**, and **Variety**, to which **Veracity**, and **Value** have been added more recently.

- That is, the huge amounts of data, the speed at which they are generated, the **different types of formats of the data**, the **trustworthiness of that data**, **and the money one can make with it.**

Networks and Social Media

- This networking capability only became possible because of major developments both in telecommunication and computer networking technologies during the **1970s**.
- But, at the same time, such changes were only made possible by new micro-electronic devices and stepped-up computing capacity.

- **Telecommunications** have been revolutionized further by the combination of “node” technologies (electronic switches and routers) and new linkages (transmission technologies).
- Major advances in **opto-electronics** (fiber optics and laser transmission) and digital packet transmission technology dramatically broadened the capacity of transmission lines.
- This **opto-electronics-based transmission** capacity, together with **advanced switching and routing architectures**, such as Transmission Control Protocol/Interconnection Protocol (TCP/IP), are the **foundation of the Internet.**

- In the **1960s** there was a call to investigate how computers could be ‘connected’ to each other in order to create an environment to enhance computer research.
- The US Department of Defense, through the Advanced Research Projects Agency (**ARPA**), created the first large computer network in **1969**.
- This resulted in **ARPANET**, which connected numerous US universities to each other. Eventually, it employed the internet protocol suite (TCP/IP) from **1983**.
- In **1988**, South Africa’s pioneering email connection to the US (and later an internet node) was set up at **Rhodes University** .

A look back…

- Further still, different forms of utilization of the radio spectrum (**traditional broadcasting**, **direct satellite broadcasting**, **microwaves**, **digital cellular telephony**), as well as coaxial cable and fiber optics, offer a diversity and versatility of transmission technologies, which are being adapted to a whole range of uses, and make possible ubiquitous communication between mobile users.
- Thus, **cellular telephony** diffused with force all over the world, taking off from the **mid-1990s**.
- In **2000**, technologies were available for a universal-coverage, personal communication device, only waiting for a number of technical, legal, and business issues to be sorted out before reaching the market.

A look back…

- From the **late 1980s** there was a growth of commercial providers of networks, i.e., the **infrastructure**.

- Together with the **WWW**, this led to the growth of the Web, where now anybody could get connected, and the so-called **New Economy of tech companies online**, such as offering the ability to buy books online rather than only in the bookshop.

Data Management toward Big Data

- While the progress of connecting devices and the Web is one of the success stories, the other one is **data management.**
- The first main step in the early **1970s** were the **relational databases** to store data and manipulate them.
- This was principally to **query the data** being stored, such as employee data, library records, and the like.

Figure 2.1: Left: Lead Apollo software engineer Margaret Hamilton standing next to the code that got the shuttle to the moon in 1969; Right: Katie Bouman, lead computer scientist, with the disks needed to store all the data and her laptop with that first image of a black hole, in 2019.

Early Data Management

- The **early 1990s** saw the **rise of data mining**: put all the data in a so-called data warehouse— a time-aware database that is slightly differently structured than a relational database— and test hypotheses on that data using association rules and statistics. For instance, with supermarket data.
- The **late 2000s** saw a third jump in the data management, combining it with even more statistical techniques and algorithms on much more data.
- Popular techniques are the various machine learning algorithms.

Machine Learning

Machine learning focuses on algorithms to achieve good predictions based on large amounts of training data.

2.2 A Time of Transition in Communication and Network Usage

- Some people claim that a new form of society has arisen through a number of major **concurrent social**, **technological**, **economic**, and **cultural transformations**.
- This change is at a global scale, but this does not entail that everyone is participating equally in it: many segments of the population of the planet are excluded from the global networks that accumulate knowledge, wealth, and power.

- A particular feature of the recent (past 20 years) **transformation** has been the radical changes in the ways **people communicate**.
- The **top-down mass media mode** has been augmented with, and to some extent replaced by, horizontal digital communication between peers and by what has been called citizen journalism.
- This brought with it a reduced amount of power by the gatekeepers of the mass media and empowered individual citizens to distribute information—or disinformation (‘**fake news’, propaganda, etc.**), as the case may be for either mode of communication.
- In addition, **wireless communication** can now reach users in most places on earth, albeit at various speeds and various cost.

- For instance, now there are **WhatsApp** groups at schools, or classes have such groups themselves for student communication, and people are used to such instant communication.
- Compare this with the '**telephone trees**' for distribution of information that was functional for many years before: a parent/student was called with a message (e.g., by the teacher), who called two other parents/students and so forth until the class was covered.
- In between, from about the **late 1990s**, and to some extent currently still, there are the email lists, there are still announcements and chatrooms to distribute information. To guess and reflect and perhaps discuss with your classmates.

Filter Bubble

- The term *filter bubble* refers to that what you see in the search results of a search engine (or feeds in social media) is determined by your prior interaction, rather than a non-personalized page (or item) rank that is not influenced by the user’s prior behavior.

Echo chamber

- The Echo chamber is defined a as a situation in which people only hear opinions of one type, or opinions that are similar to their own.

Globalization

- Globalization is a process of **interaction** and **integration** among the people, companies, and governments of different nations
- A process driven by international trade and investment and aided by information technology.
- This process has effects on the **environment**, on **culture**, on **political systems**, on **economic development** and **prosperity**, and on **human physical well-being** in societies around the world.

Localization

- **Localization** is not the opposite of the term ‘Globalization’: it is typically used in the context of localization of software.
- This concerns, mainly, the process of translation of terms and pop-up boxes of an application’s interface into the language spoken where that software is used, and adapting other features, such as spelling and grammar checking for one’s language and autocomplete for words in one’s language.
- One can also localize computer hardware, notably keyboards, adapters, and plugs.

End of presentation…

In-class Activity

- Work in groups of 3 students
- Produce one written statement stating and justifying the group’s answer to the questions
- Make a PPT or Canva presentation for presentation to the class
- Presentation shall be by group or team

Activity 1: Answer the following questions. Cite sources if possible.

1. How do you think that these different modes of communication have affected society?
2. Could you argue that one way of communicating is better than the other? If so, how is it ‘better’, or even ‘best’?
3. Is there a generational divide regarding the use of digital communication technologies?
4. If so, where about (which year) would you put the before-and-after?

Activity 2: Answer the following questions.

1. What “free services” are you using? Do you know what those companies do with your data?
2. Is it ethical to relegate individual data to being a commodity to be used by a company at their will?
3. If you build a tool that requires a user’s personal data, would you see any problems in using the data provided, be this for the purpose it was intended for or any possible purpose?
4. Some Big Data analysis may outperform humans, e.g., in recognizing tumors in brain scan images or in predicting whether a crime suspect is a criminal, yet is still not 100% correct. Should such software be used?